

Table 2. Multi-Task CGCNN Quantitative Evaluation Across 7 Core Target Properties

Property	Unit	Train MAE	Train R2	Val MAE	Val R2	Test MAE	Test RMSE	Test R2
band_gap	eV	0.092	0.992	0.066	0.995	0.070	0.096	0.996
formation_energy	eV/atom	0.065	0.993	0.069	0.984	0.063	0.083	0.994
energy_hull	eV/atom	0.068	0.993	0.069	0.991	0.076	0.093	0.994
bulk_modulus	GPa	4.680	0.991	3.719	0.991	6.603	7.370	0.888
shear_modulus	GPa	3.983	0.992	5.172	0.953	4.668	5.702	0.608
young_modulus	GPa	8.565	0.993	5.423	0.992	6.839	8.891	0.833
E_ads_DFT_eV	eV	0.083	0.989	0.096	0.994	0.055	0.063	0.995